



Minia National University
Faculty Computers & Artificial Intelligence
Final Exam: Summer 2025

Department : Computer Science
Course Title : Artificial Intelligence
Course Code : ACS 361
Lecturer : Dr. Alaa Zaki

Program: Artificial Intelligence
Date: 06 – 06 – 2025
Time: 2 Hours
Total Marks: 60 Marks

Answer the following questions

Question 1: True or False

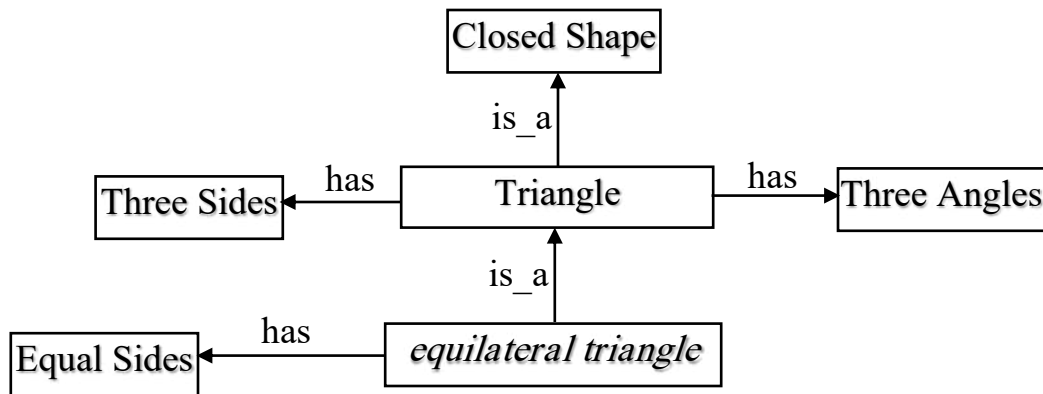
[20 Marks]

- 1- Support Vector Machines (SVM) can only be used for linear classification. ✗
- 2- Clustering algorithms like K-Means require labeled training data. ✗
- 3- A confusion matrix is used to evaluate regression models. ✗
- 4- Accuracy is the best metric for evaluating imbalanced datasets. ✗
- 5- In Machine learning model, humans write explicit rules. ✗
- 6- Machine Learning (ML) is a subset of Artificial Intelligence (AI). ✓
- 7- Supervised learning requires labeled data for training. ✓
- 8- Label encoding assigns arbitrary numbers to categories (e.g., "male"=1, "female"=2). ✓
- 9- Feature selection removes irrelevant features, while feature extraction creates new ones. ✓
- 10- Traditional programming is more suitable for tasks with fixed, unchanging rules. ✓
- 11- The training set is a collection of data used in machine learning tasks. ✗
- 12- Uniform-cost search will never expand more nodes than A* search. ✗
- 13- Email spam detection (classifying emails as "spam" or "not spam") is an example of unsupervised learning. ✗
- 14- Predicting house prices based on features like location and size is a regression task in unsupervised learning. ✗
- 15- If Cost Function = ∞ , the model perfectly predicts all training examples. ✗
- 16- Data augmentation can help reduce overfitting in deep learning models. ✓
- 17- One epoch means that the model has seen every training sample exactly once. ✓
- 18- Fill in missing values and detect and eliminate causes of dataset exceptions called data cleansing. ✓
- 19- Overfitting occurs when the model or the algorithm does not fit the data well enough. ✓
- 20- Machine learning can be used when the Rules are complex or cannot be described. ✓

Question 2: [12 Marks]

A) Represent the following facts about triangles using a *semantic network*: [4 Marks]

An equilateral triangle is a triangle whose sides are equal. A triangle is a closed shape that has three sides and three angles.



B) Translate the following sentences into predicate calculus expressions: [8 Marks]

1. *Anyone passing his history exams and winning lottery is happy.*
2. *But anyone who studies or is lucky can pass all history exams.*
3. *Omar did not study but he is lucky.*
4. *Anyone who is lucky wins the lottery.*

1. $\forall X(\text{person}(X) \wedge \text{passing}(X, \text{history_exam}) \wedge \text{winning}(X, \text{lottery}) \longrightarrow \text{happy}(X))$
2. $\forall X(\text{person}(X) \wedge \text{study}(X) \vee \text{lucky}(X) \longrightarrow \text{passing}(X, \text{history_exam}))$
3. $\neg \text{Study}(\text{omar}) \wedge \text{Lucky}(\text{omar})$
4. $\forall X(\text{person}(X) \wedge \text{lucky}(x) \longrightarrow \text{winning}(X, \text{lottery}))$

Question 3: [13 Marks]

For the following training set, a small sample of different students' performances. Here each row is one training example. Recall that in linear regression, our hypothesis is $h_\theta(x) = \theta_0 + \theta_1 x$, and the cost function

was $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_\theta(x^i) - y^i)^2$

X	Y
5	2
2	4
3	5
2	3
1	2
2	4

answer the following:

1-what is the value of m? [2 Marks]

$$M = 6$$

2-What is $J(3,2)$? [7 Marks]

$$\begin{aligned}
 J(3,2) &= \frac{1}{12} ((13 - 2)^2 + (7 - 4)^2 + (9 - 5)^2 + (7 - 3)^2 + (5 - 2)^2 + (7 - 4)^2) \\
 &= \frac{1}{12} ((11)^2 + (3)^2 + (4)^2 + (4)^2 + (3)^2 + (3)^2) \\
 &= \frac{1}{12} (121 + 9 + 16 + 16 + 9 + 9) = 15
 \end{aligned}$$

3-Suppose we set $\theta_0 = -2$, $\theta_1 = 5$. What is $h_\theta(7)$? [4 Marks]

$$h_\theta(7) = -2 + 5 \times 7 = 33$$

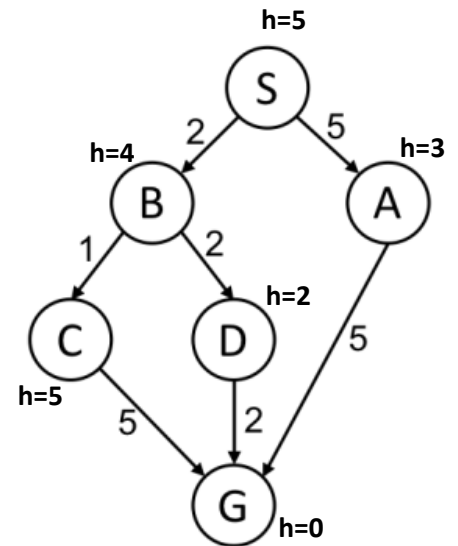
Question 4: [15 Marks]

Consider the search problem below with Start State S and Goal State G.

Edges are annotated with their cost, and stats are annotated with the heuristic values. What is the final path for this search If we use:

A) Uniform cost. [7 Marks]

B) A* search. [8 Marks]



**What is the final path for this search If we use
Uniform-Cost Search**

#	S	Frontier	Expand
0	S	S	NULL
1	B	(A,5), (B,2)	S
2	C	(A,5), (C,3), (D,4)	S, B
3	D	(A,5), (D,4), (G,8)	S, B, C
4	G,6	(A,5), (G,8,6)	S, B, C, D
Final Path is S – B – D – G, Cost = 6			

What is the final path for this search If we use A* search

#	S	Frontier	Expand
0	S	S	
1	B	(A, F=8), (B, F=6)	S
2	D	(A, F=8), (C, F=8), (D, F=6)	S, B
3	G	(A, F=8), (C, F=8), (G,6)	S, B, D
Final Path is S – B – D – G, Cost = 6			

With My Best Wishes